

Pre-MLP Norm

pre_mlp_layernorm

Router

route

Latent Down

fc1_latent_proj

Dispatch

dispatch_preprocess + dispatch

Expert MLP

routed_experts_compute

Combine

token_combine

Latent Up

fc2_latent_proj

Normalize activations
 $[T \times H]$

Project scores;
select top- K

Latent: $H \rightarrow H_{\text{latent}}$
Non-latent: 0

Permute routes;
exchange inputs

Run local MLPs
on received tokens

Exchange outputs;
unpermute and sum

Latent: $H_{\text{latent}} \rightarrow H$
Non-latent: 0

NORM

$\mathcal{O}(TH)$

GEMM + METADATA

$\mathcal{O}(THE) + \mathcal{O}(TE)$

GEMM

$\mathcal{O}(THH_{\text{latent}})$

COMM + META

$\mathcal{O}(TKH_{\text{latent}})$

GEMM

$\mathcal{O}(TKH_{\text{latent}}H_{\text{expert}})$

COMM + META

$\mathcal{O}(TKH_{\text{latent}})$

GEMM

$\mathcal{O}(THH_{\text{latent}})$

